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Article in *European Journal of Applied Physiology and Occupational Physiology* · November 1993

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Ergometric and metabolic adaptation to a 5-s sprint training programme

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Accepted June 23, 1993

Summary. The effects of 7 weeks of sprint training (repeated 5-s all-out sprints) on maximal power output ($\dot{W}_{v, \max}$) determined during a force-velocity test and a 30-s Wingate test ($\dot{W}_{\text{peak}}$) were studied in ten students [22 (SD 2) years] exercising on a cycle ergometer. Before and after training, muscle biopsies were taken from vastus lateralis muscle at rest for the ten subjects and immediately after a training session for five of them. Sprint training induced an improvement both in peak performances by 25% ($\dot{W}_{v, \max}$ and $\dot{W}_{\text{peak}}$) and in the 30-s total work by 16%. Before sprint training, the velocity reached with no load (v_0) was related to the resting muscle phosphocreatine (PCr) stores ($r=0.87$, $P<0.001$). The training-induced changes in v_0 were observed only when these PCr stores were lowest. This pointed to a possible limiting role of low PCr concentrations in the ability to reach a high velocity. The improvement in performances was linked to an increase in the energy production from anaerobic glycolysis. This result was suggested in muscle by the increase in lactate production measured after a training session associated with the 20% higher activity of both phosphofructokinase and lactate dehydrogenase. The sprint training also increased the proportion of slow twitch fibres closely related to the decrease in fast twitch b fibres. This result would appear to demonstrate an appropriate adaptive reaction following high-intensity intermittent training for the slow twitch fibres which exhibit a greater oxidative capacity.

Key words: Human skeletal muscle – Phosphocreatine – Lactate – Glycolytic enzymes – Muscle fibre conversion – Performance

Introduction

Training programme including (Nevill et al. 1989), or consisting exclusively (Thorstensson et al. 1975) in short periods of intense exercise have been shown to increase peak power production ($\dot{W}_{\text{peak}}$) during sprint exercises. However, the metabolic mechanisms underlying this enhancement have not been elucidated. Intense exercise lasting a few seconds has been found to be mainly dependent on anaerobic processes (Hirvonen et al. 1987; Medbø et al. 1989). Sprint training has been shown to induce no change in the high-energy phosphate [adenosine 5' triphosphate (ATP); phosphocreatine (PCr)] stores in resting muscle (Rehunen et al. 1982; Thorstensson et al. 1975). On the other hand, the depletion of the PCr stores during a running session was greater in the faster subjects as has been demonstrated in a national top level group of sprinters (Hirvonen et al. 1987). Insignificant effects of sprint training on muscle glycolytic activity, as expressed by lactic dehydrogenase (LDH) activity, have been reported (Thorstensson et al. 1975). On the other hand, higher muscle LDH activity has been measured in sprint athletes, than in sedentary subjects (Costill et al. 1976). Furthermore, the relationships between $\dot{W}_{\text{peak}}$ and anaerobic energy release remain unclear. Some findings (Thorstensson et al. 1975; Boobis et al. 1983) have suggested that $\dot{W}_{\text{peak}}$ may be increased while anaerobic energy release remains unchanged. On the other hand, similar increases in high anaerobic energy release defined as accumulated O_2 deficit and peak rate of anaerobic energy release have been obtained by training sessions lasting either 20 s or 2 min (Medbø and Burgers 1990). As intense exercise is dependent on anaerobic processes, high-intensity intermittent work requires aerobic metabolism during the rest periods for the resynthesis of depleted PCr and the removal of lactate accumulated during the period of exercise. Thus, Simoneau et al. (1985) have shown that high-intensity intermittent training induced an increase in the proportion of slow twitch (ST) fibres associated with a concomitant decrease in the fast twitch b (FTb) fibres.

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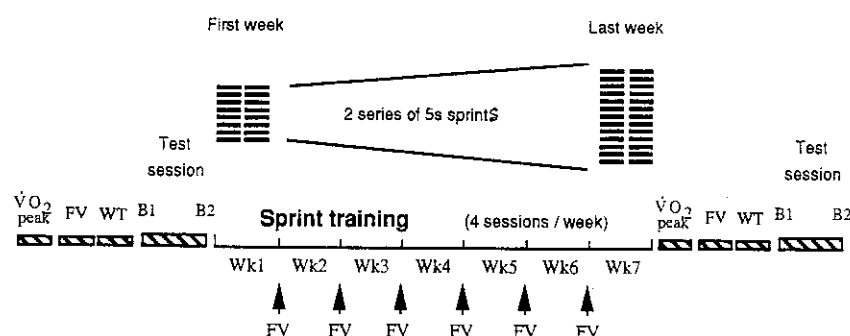


Fig. 1. Experimental design of a training programme with different tests. FV, force-velocity relationship; WT, 30-s Wingate test; Test session, a training session consisting in eight sprints to 80% maximal force B_1 and B_2 , muscle samples at rest and after the completion of the test session, respectively: $\dot{V}O_{2\text{peak}}$, peak oxygen uptake

The purpose of the present study was therefore to investigate the effects of high-intensity intermittent training consisting exclusively of short sprints (5 s) on the performance obtained both in this specific exercise and in exercises of 30-s duration in relation to aerobic and anaerobic adaptations in the muscle involved.

Methods

Subjects. Ten students (eight men and two women) aged 22 (SD 2) years [height, 170 (SD 9) cm; body mass, 63.7 (SD 8.5) kg; peak oxygen uptake ($\dot{V}O_{2\text{peak}}$): 51.2 (SD 7.3) $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$] volunteered for this study. Each subject was informed of the purpose, possible risks and benefits of the study prior to giving written consent to participate. The protocol was approved by the Ethics Committee of Saint-Etienne Teaching Hospital.

Protocol. The subjects were moderately active before the study. They were first familiarized with the operation of the cycle ergometer and the performance of the different tests. The schedule of tests and of the training sessions is shown in Fig. 1. All the tests were performed in the morning. First the $\dot{V}O_{2\text{peak}}$ of the subjects was measured. Then they were submitted on different days to measurements of the force-velocity relationship, a Wingate test and a test session before and after sprint training. The postsprint training test started 2 days after the last training session had been completed.

Ergometric tests. Oxygen uptake was measured by the Douglas bag method during a progressive exercise test (37-W increments) with steps of 4-min duration separated by 1-min rests until voluntary exhaustion. The highest value was considered as $\dot{V}O_{2\text{peak}}$.

Force-velocity and Wingate tests were conducted with an Ergomeca cycle ergometer, equipped with an arm balance allowing braking forces up to 200 N (Brue 1982). One pedal revolution was equivalent to a 6-m progression at the perimeter of the flywheel. The subjects had to stay seated on the saddle throughout a sprint and their feet were strapped to the pedals to prevent them from slipping.

The force-velocity relationship was obtained using the procedure described by Vandewalle et al. (1987). This test consisted of short maximal sprints against increasing braking forces. After a warm-up, the subjects performed two or three sprints against low braking forces. They then rested for 5 min. The test began with a braking force equal to 10 N. The maximal peak velocity was reached in 4–8 s. After a 5-min recovery period, the braking force was increased by 10 N. The test was stopped when there was no longer any increase in the power (calculated from the peak velocity and the braking force). The maximal velocity was also measured when pedalling with no load (v_0). The linear force-velocity and parabolic force-power relationships were assessed (Fig. 2). The braking force for which the maximal power output ($W_{v,\text{max}}$)

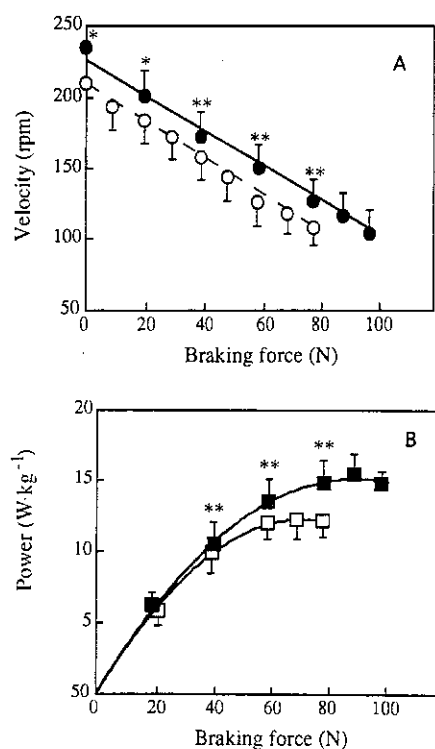


Fig. 2. Effects of sprint training on force-velocity (A) and force-power (B) relationships. * $P < 0.05$; ** $P < 0.01$. A ○ Pre-training; ● post-training. B □ Pre-training; ■ post-training

was obtained at maximal velocity, corresponded to the optimal force (F_{max}). The $W_{v,\text{max}}$, which defined the power corresponding to maximal velocity of pedalling, did not take into account the inertia of the flywheel. It did not therefore directly express maximal anaerobic power sustained by the subjects (Lakomy 1986).

The Wingate 30-s test was performed at 75% F_{max} , using the procedure described by Bar-Or (1987), after a 10-min warm up at 50% $\dot{V}O_{2\text{peak}}$ interspersed by 3 to 4 sprints. The flywheel velocity was monitored every 0.5 s and displayed by the computer MEV 2000 (Mazet Electronic, Le Mazet St Voy) after the completion of the test. Four parameters were determined: W_{peak} , which referred to the power corresponding to the maximal velocity reached during the test; t_{peak} , the time necessary to reach this maximal velocity; W_{30} , the power measured at the 30th s of the test; W_{tot} , the total work performed during the test, estimated as the integral of the power throughout the time period of the test (Fig. 3).

The test session consisted in two series of 8 intermittent sprints (5-s sprint, 55-s rest) at 80% F_{max} . The recovery period between the two series was 15 min.

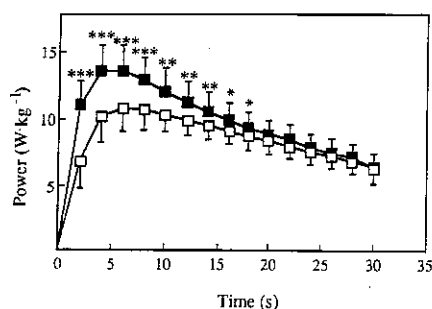


Fig. 3. Power output during a 30-s maximal sprint on a cycle-ergometer before and after 7 weeks of sprint training ($n=10$). □ Pre-training; ■ post-training

Sprint training. The sprint training programme lasted for 7 weeks at the rate of four sessions each week. Each session consisted of two series of maximal intermittent sprint cycling (5-s sprint, 55-s rest) at 80% $F_{\max}$ (i.e. 130–140 rpm). The rest period between the two series was 15 min. The first week sessions included 8 sprints in each series (see Fig. 1). This number was increased by 1 sprint every week so that each series of the last weeks sessions included 13 sprints. The force-velocity relationship was established every week before a training session to adjust the braking force so that the subjects were always training at an intensity equivalent to 80% $F_{\max}$.

Muscle sampling. Small incisions of skin and fascia were made under local anaesthesia to take biopsies from vastus lateralis muscle with Weil Blakesley forceps (Henriksson 1979). The mean mass of a muscle sample was estimated to be 100 mg wet mass.

At rest, the muscle samples were taken for all of the subjects ($n=10$) before the test session at the beginning and at the end of the training programme. These were large enough to allow both histochemical analysis and measurements of the different metabolites and enzyme activities. A part of these samples was directly frozen and stored in liquid nitrogen until analysed for metabolites and enzymes. The other part was mounted in Tissue-Tek II O.C.T. Compound (Miles Lab. Inc. Ill., USA) for histochemical analysis before freezing in isopentane cooled in liquid nitrogen. The interval between removal of a muscle biopsy sample and freezing was 2–4 min.

At the end of the test, the muscle samples were taken from only five of the subjects. They were directly frozen and stored in liquid nitrogen until analysed for metabolites. The time taken from the end of the sprint to immersion of the samples in liquid nitrogen averaged 17.5 (SD 5.0) s.

Thus five subjects had four biopsies whereas the other five subjects only had two biopsies.

Histochemical analysis. Cryostat serial transverse sections (-20°C , 10 μm) were stained for adenosine triphosphatase activity (ATPase) to determine the muscle fibre composition (Brooke and Kaiser 1970) and for nicotinamide adenine dinucleotide tetrazolium reductase activity (Novikoff et al. 1961) to measure the mean cross-sectional area of the FT and ST fibres. For each subject, at least 709 (SD 188) fibres were classified. Mean muscle fibre areas were determined by averaging the cross-sectional areas of 20 randomly selected fibres of each type from ATPase stained reaction that had been photographically reproduced at a known magnification. The fibre composition is expressed relative to the mean cross-sectional area.

Muscle metabolites. The muscle sample was freeze dried, dissected free of connective tissue and blood, weighed, extracted with 0.65 mol·l $^{-1}$ HClO $_4$ and centrifuged at 4 $^{\circ}\text{C}$ and 600 g for 5 min. The supernatant (muscle extract) was neutralized and as-

sayed enzymatically in triplicate for creatine, PCr, ATP, glucose 6 phosphate (G6P) and lactate using fluorimetric procedures of Lo-wry and Passonneau (1972).

Enzyme activities. The muscle sample was weighed at -20°C and manually homogenized at 4 $^{\circ}\text{C}$ in 0.1 mol·l $^{-1}$ phosphate buffer (pH=8.2) containing 5 mmol·l $^{-1}$ 2-mercaptoethanol, 30 mmol·l $^{-1}$ NaF, 5 mmol·l $^{-1}$ MgCl $_2$ and 0.5 mmol·l $^{-1}$ ATP. This tissue suspension was immediately used to measure spectrophotometrically phosphofructokinase [PFK Enzyme Commission no. (EC) 2.7.1.11] activity using the method of Mansour (1966), then frozen and stored at -80°C until analysed for the other enzymes. The activities of hexokinase (EC 2.7.1.1), lactate dehydrogenase (LDH, EC 1.1.1.27), citrate synthase (EC 4.1.3.7) and 3-hydroxyl acyl-CoA dehydrogenase (EC 1.1.1.35) were determined fluorimetrically (Essen-Gustavsson and Henriksson 1984). All the enzyme activities were measured at 25 $^{\circ}\text{C}$.

Anthropometric analysis. The cross-sectional area of the vastus lateralis muscle was measured planimetrically (digitalizing tablet Numonics 2200) on computed tomography radiographs (CGR CE 10000) of both thighs at midthigh level, i.e. at the midpoint between the trochanter major and the articular cleft between the femoral and tibial condyles (Schantz et al. 1983).

Expression of the results. The muscle metabolite concentrations are expressed with respect to total creatine except for lactate which is expressed in millimoles per kilogram muscle dry mass, the enzyme activities in micromoles per minute per gram wet mass. The adjustment to total creatine content allowed correction for variability in solid nonmuscle constituents between biopsies.

Statistical procedures. The statistical significance of the differences between two mean values was assessed by Wilcoxon rank-sum test for paired values. Correlation coefficients were calculated using the least squares method.

Results

Performance changes

These are listed in Table 1. The $\dot{V}O_{2\text{peak}}$ remained unchanged. The mean training power output increased regularly exceeding its initial value by 17%, when the training was completed (Fig. 4).

The force-velocity relationship was shifted so that the velocity at every load including v_0 was higher after than before sprint training (Fig. 2A). These changes resulted in an increase of both $F_{\max}$ and $\dot{W}_{v,\max}$ by 28% and 23% respectively (Fig. 2B).

The Wingate test performances were also improved (Fig. 3): $\dot{W}_{\text{peak}}$ and $\dot{W}_{\text{tot}}$ were significantly increased by 26% and 16% respectively, while t_{peak} was decreased by 20%. The power output calculated every 2 s was significantly higher up to 18 s. The improvement in $\dot{W}_{\text{peak}}$ was significantly related to that of v_0 and t_{peak} ($r=0.65$, $P<0.05$ and $r=-0.63$, $P<0.06$, respectively). $\dot{W}_{30}$ was unaltered.

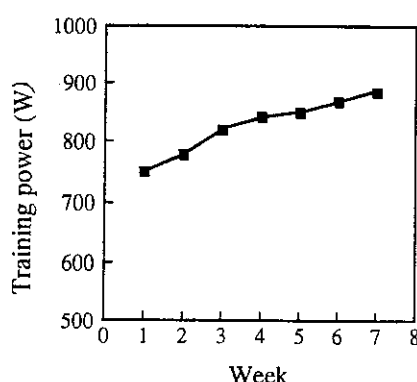
Muscle metabolic changes

At rest, PFK and LDH activities were increased by 19% and 20%, respectively, while oxidative enzyme activities remained unaltered (Table 2). Total creatine

Table 1. Physical characteristics and performance variables measured before and after sprint training ($n=10$)

	Pre-training		Post-training		Significance
	Mean	SD	Mean	SD	
$\dot{V}O_{2\text{peak}}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	51.2	7.3	50.8	6.9	ns
Training					
F_{training} (N)	52.0	9.8	67.7	13.7	$P<0.001$
W_{training} ($\text{W} \cdot \text{kg}^{-1}$)	11.6	1.6	13.6	2.0	$P=0.005$
Force-velocity test					
v_0 (rpm)	208	36	235	15	$P<0.05$
F_{max} (N)	65.7	11.8	84.9	16.0	$P<0.005$
$W_{v,\text{max}}$ ($\text{W} \cdot \text{kg}^{-1}$)	12.2	1.6	15.2	2.1	$P=0.005$
Wingate test					
F_{WT} (N)	49.2	9.9	62.5	12.9	$P<0.001$
W_{peak} ($\text{W} \cdot \text{kg}^{-1}$)	10.9	1.6	13.7	2.0	$P=0.005$
t_{peak} (s)	5.0	1.3	4.0	0.6	$P<0.05$
W_{30} ($\text{W} \cdot \text{kg}^{-1}$)	6.3	0.8	6.2	1.1	ns
W_{tot} (kJ)	16.5	2.9	19.1	3.1	$P<0.01$

$\dot{V}O_{2,\text{peak}}$, peak oxygen uptake; F_{training} , braking force during training; W_{training} , training power output; v_0 , maximal velocity with no load; F_{max} , optimal force; $W_{v,\text{max}}$, maximal power output; F_{WT} , force during test; W_{peak} , power at maximal velocity; t_{peak} , time to reach maximal velocity; W_{30} , power at 30th s of test; W_{tot} , total work during test

**Fig. 4.** Changes in power produced by training**Table 2.** Muscle enzyme activities before and after 7 weeks of sprint training ($n=10$)

	Pretraining ($\mu\text{mol} \cdot \text{mn}^{-1} \cdot \text{g}^{-1}$ wet mass)		Posttraining		Significance
	Mean	SD	Mean	SD	
Citrate synthase	6.4	1.7	7.3	0.9	ns
Hydroxyl acyl-CoA dehydrogenase	8.8	2.3	10.0	1.9	ns
Hexokinase	2.2	0.5	2.5	0.4	ns
Phosphofructokinase	45.2	11.3	53.8	7.1	$P<0.05$
Lactate dehydrogenase	236	89	285	96	$P<0.05$

was constant before and after training [before 116 (SD 21 mmol, after 111 (SD 13) mmol, ns]. Resting mean PCr [before 0.62 (SD 0.06), after 0.65 (SD 0.03),

Table 3. Muscle metabolite changes during a training session before and after 7 weeks of sprint training ($n=5$)

	Pretraining		Posttraining		Significance
	Mean	SD	Mean	SD	
Δ Phosphocreatine	-0.19	0.07	-0.20	0.02	ns
Δ Adenosine 5' triphosphate	-0.002	0.020	-0.030	0.020	ns
Δ Glucose 6 phosphate	0.022	0.020	0.035	0.016	ns
Δ Lactate	37.2	17.9	52.8	13.5	$P<0.005$

Results are expressed with respect to total creatine except for lactate which is expressed in millmoles per kilogram of dry mass

Table 4. Morphological characteristics before and after 7 weeks of sprint training ($n=10$)

	Pre-training		Post-training		Significance
	Mean	SD	Mean	SD	
Mass (kg)	63.7	8.5	64.6	8.8	$P<0.05$
Left muscle area (mm^2)	4534	1088	4658	1013	ns
Right muscle area (mm^2)	4534	1097	4492	934	ns
% Slow twitch fibres	43	13	51	11	$P<0.05$
% Fast twitch fibres-a	41.6	11.3	41.5	10.3	ns
% Fast twitch fibres-b	15.3	7.7	6.8	7.6	$P<0.01$
Slow twitch fibre area (μm^2)	5640	1840	5040	1370	$P<0.05$
Fast twitch fibre area (μm^2)	5480	1540	5140	1200	ns
% Slow twitch fibre area	44	14	51	12	$P<0.1$

ns] and ATP [before 0.17 (SD 0.01), after 0.18 (SD 0.03), ns] remained unchanged.

During the sprint series, the PCr depletion was unchanged by the training whereas the increases in lactate concentration were higher after than before training with a similar G6P accumulation (Table 3).

It is noteworthy that %ST was increased and ST area was decreased resulting in an unchanged %ST area. The increase in %ST fibres was related to the decrease in %FTb fibres ($r=0.77$, $P<0.01$). These results were not associated with any change in cross-sectional muscle area (Table 4).

Before sprint training, the ergometric parameters such as $W_{v,\text{max}}$, W_{peak} or v_0 were correlated with resting muscle concentration of PCr ($r=0.92$, $r=0.85$ and $r=0.87$, $P<0.01$, respectively). Sprint training induced some large improvements in PCr only in the four subjects who had a low PCr concentration before training, thus these relationships were not obtained after training (Fig. 5).

Discussion

Performance changes

The findings of this study demonstrated that a training programme consisting exclusively in short sprints induced a substantial improvement both in sprint W_{peak}

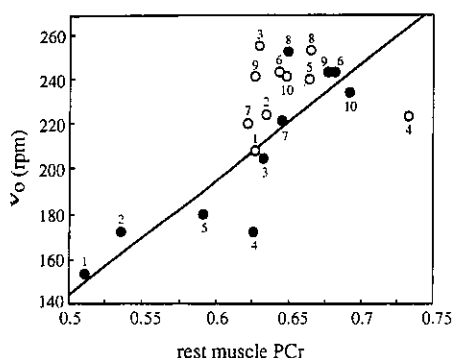


Fig. 5. Relationship between the velocity (v_0) and the resting muscle phosphocreatine (PCr) stores before and after 7 weeks of sprint training ($n=10$). The relationship is significant only before training (●). The post-training values (○) were included to illustrate the grouping values. The PCr values are expressed with respect to total creatine content. ○ Post-training ns; ● pre-training. $y=515 \cdot x - 113$; $r=0.87$, $P<0.001$

and $\dot{W}_{v, \max}$ and in W_{tot} performed during 30-s maximal exercise. These improvements were found for all the subjects regardless of sex. The two female subjects showed, before training, $\dot{W}_{\text{peak}}$ of 8.4 and 9.5 $\text{W} \cdot \text{kg}^{-1}$, respectively, compared to 11.4 (SD 1.4) $\text{W} \cdot \text{kg}^{-1}$ in the men. This difference is in agreement with the data of Brooks et al. (1990). In spite of these differences, the increases in performances were similar in both sexes [3.0 and 1.7 $\text{W} \cdot \text{kg}^{-1}$ in the women; 2.9 (SD 1.8) $\text{W} \cdot \text{kg}^{-1}$ in the men].

The higher $\dot{W}_{\text{peak}}$ and $\dot{W}_{v, \max}$ at the end of the programme were the result of a progressive increase in the power during the training (see Fig. 4). This improvement in $\dot{W}_{\text{peak}}$ and $\dot{W}_{v, \max}$ induced by short sprint training has already been demonstrated by Thorstensson et al. 1975, Costill et al. 1979, and Nevill et al. 1989. The alteration in the force-velocity relationship is consistent with the results of cross-sectional studies (Vandewalle et al. 1987); before training, the ergometric characteristics of our subjects were similar to those of recreational subjects and the sprint training programme shifted their ergometric performances towards the pattern of ball game players whose activity includes mainly intermittent sprints. The decrease in t_{peak} associated with the improvement of $\dot{W}_{\text{peak}}$ would suggest that the flywheel was more vigorously accelerated. The improvement in $\dot{W}_{\text{peak}}$ was thus even greater than expressed by the increase of $\dot{W}_{v, \max}$.

The improvement limited to the power output during the first 18 s of exercise would seem to be in agreement with previous studies. Thus, a sprint training programme including some 6-s sprints but mainly consisting in 30-s and 2-min supramaximal exercises induced an increase in sprinting $\dot{W}_{\text{peak}}$ without significant improvement in the W_{tot} performed during a 30-s sprint (Nevill et al. 1989). On the other hand, an increase in the W_{tot} during the 30-s Wingate test, induced by sprint training on a treadmill or on a cycle ergometer, had never been demonstrated before. Denis et al. (1992) have pointed out that the ability of two groups to per-

form a similar W_{tot} during a 45-s Wingate test revealed two distinct mechanical profiles, i.e. higher instantaneous power during the first 20 s and a more pronounced fatigability in 100-m runners than in 800-m runners. This different pattern of energy output during supramaximal exercise was explained by specific metabolic profiles in each group. Thus, the ability to produce high power during a few seconds was related more to anaerobic metabolism whereas the ability to sustain supramaximal work lasting 30–45 s was related more to oxidative capacity of the muscles.

Muscle metabolism changes

The improvement in $\dot{W}_{\text{peak}}$ and $\dot{W}_{v, \max}$ was not related to a significant increase in mean resting muscle PCr concentration. This result is in line with data of both cross-sectional studies on athletes (Rehunen et al. 1982; Denis et al. 1992) and longitudinal observations of the effects of short sprint training (Thorstensson et al. 1975). However, a relationship was established before training between v_0 and PCr stores at rest. Moreover, the large improvements in v_0 were associated with an increase in the PCr stores only when these stores were very low (<0.625 , i.e. 70 $\text{mmol} \cdot \text{kg}^{-1}$ dry mass). These observations point to a possible limiting role of the PCr concentration at rest in the ability to reach a high velocity when the subjects were not metabolically adapted, the PCr being the most quickly released energy. This limiting role had not been demonstrated by previous studies (Rehunen et al. 1982; Thorstensson et al. 1975) because of the high PCr concentration at rest in the subjects studied who were either highly-trained athletes or moderately trained subjects. Moreover, in our study, this relationship disappeared with training showing the minor importance of resting PCr concentration at higher physical training levels (Fig. 5). On the other hand, the rate of the PCr-creatine shuttle could be greater in the achievement and in the repetitions of $\dot{W}_{\text{peak}}$ and $\dot{W}_{v, \max}$. An increase in the activities of the high energy phosphate transferring enzymes with sprint training as observed by Thorstensson et al. (1975) could explain the relationship between the improvements in $\dot{W}_{\text{peak}}$ and the shortening of t_{peak} . This is consistent with the greater depletion of the PCr stores during running in the faster subjects as has been indicated in a homogeneous group of sprint runners by Hirvonen et al. (1987). Furthermore, an increase in these enzyme activities should allow a higher PCr re-synthesis during the rest periods. This could explain the similar PCr depletion in spite of a higher $\dot{W}_{\text{peak}}$ reached at every session after than before sprint training.

Another finding of our study was the improvement in $\dot{W}_{\text{peak}}$ and $\dot{W}_{v, \max}$ associated with an increase in the energy production from anaerobic glycolysis as demonstrated by Nevill et al. (1989). This was supported, in our study, by a higher muscle lactate concentration measured after the completion of the sprint series in the post-training state. It has been shown that, after a

high-intensity exercise resulting in the production of the same W_{tot} , sprint-trained athletes has higher muscle and blood lactate concentrations than middle-distance trained athletes (Denis et al. 1992). In that study, the lactate accumulation was closely related to the ability to produce high power in a few seconds ($r=0.87$, $P<0.001$). This result is in agreement with earlier reports establishing that muscle lactate increases rapidly right from the start of supramaximal exercise (Jacobs et al. 1983; Hirvonen et al. 1987). This enhanced involvement of glycolytic processes was further demonstrated, in our study, by the increased muscle PFK and LDH activities. Increased PFK activity as a result of sprint training is a common finding (Sharp et al. 1986; Cadefau et al. 1990). In contrast, these studies did not reveal any increase in muscle LDH content. On the other hand, a cross-sectional study has demonstrated that sprint athletes have a higher LDH content than do long-distance runners (Costill et al. 1976). In fact, the LDH content includes both the iso-enzyme fractions that participate in pyruvate reduction and in lactate oxidation; opposite variations of these two groups of LDH iso-enzymes have been observed during a sprint training season (Cadefau et al. 1990). Discrepancies between studies dealing with sprint training could be related to the duration of training and the exercise periods during the sprint series.

The significant increase in %ST as a result of short sprint training seems paradoxical because sprint-trained athletes generally have a high FT percentage (Costill et al. 1976) and some increases in FT fibres have been reported after sprint training (Jansson et al. 1990). However, these latter studies have included periods of training exercise lasting at least 30 s in which energy production relied more upon anaerobic capacity. In our study, the increase in ST fibres was associated with a decrease in FTb fibres. Cadefau et al. (1990) have shown that 8 months of a specific sprint training programme induced an increase in the percentage of ST fibres in young athletes. Furthermore, Simoneau et al. (1985) have observed that high-intensity intermittent training induced a higher proportion of ST fibres, when expressed as both percentage and percentage area, with a concomitant decrease in FTb fibres, when compared to a control group in which no significant difference was observed. In our study, we had no control group. However, the sampling and measuring procedures were standardized as far as possible: all the biopsies were taken by the same person; at rest, the pre- and post-training muscle samples were made on the same thigh, 2 or 3 cm distant from each other, moreover, for each subject, the pre- and post-training histochemical sections were stained simultaneously. Thus, the fibre changes observed in our study should be the result of the specific sprint training and not the result of the variability of the muscle biopsy or staining techniques.

Conflicting evidence has been reported regarding the adaptability of fibre type areas to exercise training. Commonly, skeletal muscle has been thought to respond to physiological overload by increasing fibre size

(Staron et al. 1989). Thus, strength training has been shown in general to induce muscle hypertrophy (Gollnick et al. 1972; Costill et al. 1976, 1979; McDougall 1986). However, strength training induced hypertrophy has been shown to result in a slowing down of the rate of tension development (Gonyea and Bonde Petersen 1978). It has often been associated with a reduced oxidative capacity (Costill et al. 1976; Gollnick et al. 1972; Tesch et al. 1984). This adaptive process might be counterproductive when a rapid rate of tension development is needed and/or when the exercise is performed intermittently. The results of the present investigation add evidence to the notion that the adaptation of the fibre type areas could depend both on the nature and duration of the training programme and on the pretraining status of the subjects.

Thus, intense and repetitive exercise could induce the conversion of intermediate fibres to the oxidative type. Because the training exercises were performed intermittently and due to their greater oxidative capacity, the ST fibres could have been more extensively involved during rest periods than the FT in the replenishment of the depleted PCr stores and the removal of the lactate accumulated during exercise periods.

In conclusion, the high-intensity intermittent training used in this study induced an improvement in performance. This type of training resulted in an adaptive response both in greater glycolytic enzyme activities and in an increased proportion of ST fibres. The former adaptation might be related to the intensity of the brief supramaximal exercise. The latter might be more the result of the repetitive intense exercise. Low PCr stores in muscle could be a factor limiting the performance.

Acknowledgements. The authors thank Prof. J. Coudert and N. Fellmann, University of Clermont-Ferrand, for their thorough discussions and comments. They also gratefully acknowledge the help of John Carew in the correction of the English manuscript.

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